Theoretical Background

Geometry

The physical domain consists of two infinite parallel plates separated by a constant distance H. The coordinate system is defined such that the x-axis is aligned with the streamwise flow direction, and the y-axis is normal to the plates.

- **Domain:** $0 \leq y \leq H$ and $-\infty < x < \infty$
- **Top Plate ($y = H$):** Stationary (velocity $U = 0$)
- **Bottom Plate ($y = 0$):** Moving with a constant velocity U_w

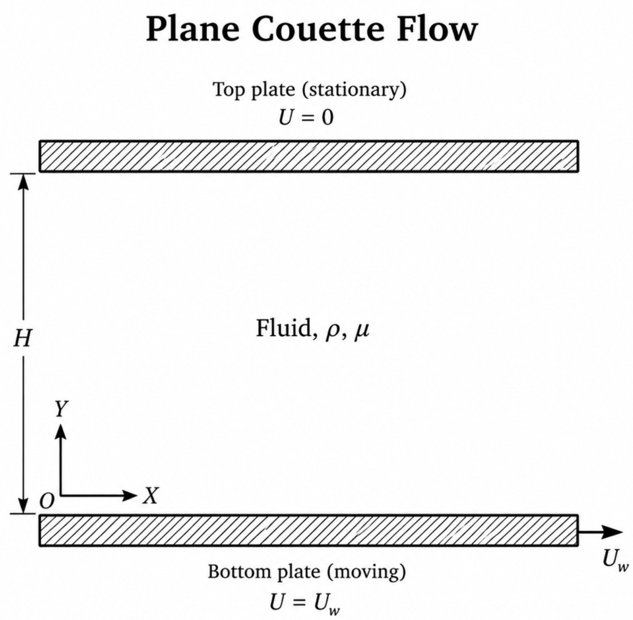

Figure 1: Channel Geometry

As illustrated in Figure 1, the fluid occupies the region between the two parallel plates, with the origin of the coordinate system located on the bottom plate.

Governing Equations

To derive the mathematical model for plane Couette flow, the following assumptions are adopted:

1. The fluid is incompressible, and its density ρ is constant.
2. The thermophysical properties of the fluid are constant, including the dynamic viscosity μ.
3. The flow is one-dimensional; therefore, the velocity has only an x-component.
4. The plates are infinitely long in the flow direction.

5. No external pressure gradient is imposed in the streamwise direction, such that $\partial p/\partial x = 0$.
6. Gravitational effects and other body forces are neglected.

Continuity Equation

The general form of the continuity equation is:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0 \tag{1}$$

For a two-dimensional Cartesian coordinate system, Equation (1) is expressed as:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} = 0 \tag{2}$$

Given that the density is constant, $\partial \rho/\partial t = 0$, Equation (2) reduces to:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{3}$$

For the one-dimensional plane Couette flow considered here, the velocity field has only a streamwise component:

$$\mathbf{V} = [u(y,t),\, 0]$$

Therefore, the velocity component normal to the plates is identically zero:

$$v = 0$$

Thus, Equation (3) simplifies to:

$$\frac{\partial u}{\partial x} = 0 \tag{4}$$

It follows from Equation (4) that the streamwise velocity is independent of x and can be written as:

$$u = u(y,t)$$

Momentum Equation

For an incompressible Newtonian fluid with body-force effects neglected, the Navier–Stokes momentum equation is:

$$\rho\left(\frac{\partial \mathbf{V}}{\partial t} + \mathbf{V}\cdot\nabla\mathbf{V}\right) = -\nabla p + \mu\nabla^2\mathbf{V} \tag{5}$$

Given the velocity field $\mathbf{V} = [u(y,t),\, 0]$, the x-momentum component becomes:

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \tag{6}$$

Substituting the kinematic conditions:

$$\frac{\partial u}{\partial x} = 0, \qquad v = 0, \qquad \frac{\partial^2 u}{\partial x^2} = 0$$

together with the zero streamwise pressure gradient:

$$\frac{\partial p}{\partial x} = 0$$

into Equation (6) yields:

$$\rho\frac{\partial u}{\partial t} = \mu\frac{\partial^2 u}{\partial y^2} \tag{7}$$

Dividing Equation (7) by the density gives the one-dimensional transient diffusion equation:

$$\frac{\partial u}{\partial t} = \nu\frac{\partial^2 u}{\partial y^2} \tag{8}$$

where:

$$\nu = \frac{\mu}{\rho}$$

is the kinematic viscosity.

Additionally, the y-momentum equation reduces to:

$$\frac{\partial p}{\partial y} = 0$$

which confirms that the pressure is uniform across the channel height.

Analytical Solution

(This section is reserved for the analytical solution comparison.)

Initial and Boundary Conditions

The initial and boundary conditions for the transient plane Couette flow problem are defined as follows.

Initial Condition

At the initial time $t = 0$, the fluid is at rest throughout the channel interior, while the bottom plate is already moving with the constant velocity U_w:

$$u(0, 0) = U_w$$

$$u(y, 0) = 0 \qquad \text{for } 0 < y \leq H$$

Boundary Conditions

For $t \geq 0$, the velocity field satisfies the following Dirichlet boundary conditions:

$$u(0, t) = U_w$$

$$u(H, t) = 0$$

The first condition represents the no-slip condition at the moving bottom plate, while the second condition represents the no-slip condition at the stationary top plate. Together, these conditions describe the impulsive start of the bottom plate and the subsequent transient diffusion of momentum across the channel.